

1:1 SCALE CUSTOM DSLR CAMERA

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Based on Original Concept by: Sc Geekware
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MODIFIED 150% SCALE BUILD GUIDE

PROJECT SUMMARY

This assembly guide details the customized 1:1 real-scale DSLR camera chassis built around an ESP32-CAM module and an integrated 2.8" SPI TFT LCD Viewfinder (ILI9341). By scaling the chassis to 150% and splitting the main body into two flat halves, the print time is reduced to ~3 hours per section while eliminating slicer geometry errors and support requirements.

STL File Handling & Slicing Protocol

File Name	Required Scaling	Action & Slicing Notes
body.stl	Scale to 150%	Resize to 150% in slicer. Split into Front and Back halves. Place flat face down for ~3hr print time without supports.
lens.stl	Scale to 150%	Resize uniformly to 150%. Slice with 3–4 wall lines to preserve mechanical strength on screw threads.
body+lens.stl	Already Perfect Size (No Scaling)	Pre-scaled combined model. Ready to import, slice directly, and generate fresh G-code for your printer.
*.gcode (Included)	Do Not Use	Targeted specifically for stock Creality Ender-3 Pro at original scale. Must re-slice raw STLs for custom setup.

Required Materials & Hardware

Qty	Item / Component	Notes / Specifications
1	ESP32-CAM Module	AI-Thinker with OV2640 Camera Lens
1	2.8" SPI TFT Display	ILI9341 Driver (240x320 resolution with SPI interface)
1	Micro USB Breakout Board	Female Micro USB to DIP Adapter
1	3D Printed Body (Front Half)	150% Scale from body.stl , cut flat face down
1	3D Printed Body (Back Half)	150% Scale from body.stl , screen cutout expanded for 2.8" display
1	3D Printed Lens Assembly	150% Scale from lens.stl with mounting frame

Qty	Item / Component	Notes / Specifications
5	M2x5 Flat Head Screws	For secure internal component mounting
3	M2x5 Button Head Screws	For sub-assembly and frame attachment
1	Ribbon / Dupont Wire Set	Red/Black 10cm wires for internal 5V/GND power routing

3D Printing Specifications

- **Scale Factor:** 150% Uniform Scaling for `body.stl` & `lens.stl` (Reaches ~130mm x 100mm x 75mm 1:1 real DSLR dimensions). `body+lens.stl` requires no scaling.
- **Body Separation:** Model split down the center into Front and Back halves.
- **Print Orientation:** Both halves printed flat-side-down onto the build plate.
- **Print Settings:** 0.16mm - 0.2mm layer height, 15–20% Infill, 3–4 Wall Perimeters. No support material needed on main seam.
- **Material:** PLA+ or PETG for structural rigidity.

Assembly & Build Steps

Step 1 Prepare USB Power Breakout & Wiring

- Cut 35mm lengths of red (5V) and black (GND) jumper wires.
- Strip the ends, apply 10mm heat shrink, and solder directly to the VBUS and GND pads of the Micro USB breakout board.
- Bend the solder joints gently to 90° for low-profile clearance inside the body bottom.

Step 2 Fit & Align ESP32-CAM Board

- Flash your ESP32-CAM code and verify camera operation before installation.
- Mount the ESP32-CAM module into the custom 1:1 inner mounting bracket/frame.
- Align the OV2640 camera lens through the central lens hole on the front body half. Tighten with an M2x5 button head screw.

Step 3 Install 2.8" SPI TFT LCD Viewfinder

- Position the 2.8" ILI9341 display into the expanded rear screen opening on the back body half.
- Wire the SPI interface pins (CS, RESET, DC, MOSI, SCK, VCC, GND) to the designated ESP32-CAM GPIO headers.
- Secure display edges using the internal mounting tabs or non-conductive adhesive.

Step 4 Power Connections & Chassis Joining

- Connect the power wires from the Micro USB breakout board to the **5V** and **GND** pins on the ESP32-CAM.
- Bring the Front and Back chassis halves together, using alignment pins/dowels along the seam.
- Fasten the chassis using M2x5 screws along bottom mounting holes or secure with cyanoacrylate adhesive.

Step 5 Lens Barrel & Dial Knob Attachment

- Press the printed lens holder onto the camera body, ensuring proper groove alignment with a spudger.
- Thread the scaled lens barrel onto the front mount snugly.
- Attach the top mode dial knob and snap cover pieces into place. Assembly is complete!

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Modified Work Title: ESP32-CAM 1:1 Scale DSLR Body with Viewfinder

Document & Revision Author: Made by Russel

Original Model Inspiration: Sc Geekware

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